



ARI SWERDLOW

PRODUCT ENGINEER / APPLIED COMPUTER VISION

Browser products where movement becomes input - built for noisy signals, real devices, and clear recovery.

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PROFILE

Independent product engineer working across computer vision, interaction design, game systems, and release reliability. I turn uncertain camera input into sessions people can understand and recover.

CORE STRENGTHS

- Computer vision product UX
- Browser AR and camera pipelines
- Finite-state game systems
- Simulation and recorded replay
- On-device performance budgets
- Accessibility and recovery design
- Release gates and observability

TECHNOLOGY

TypeScript, React, Next.js, Three.js, React Three Fiber, MediaPipe, TensorFlow.js, WebGL, Python, FastAPI, Playwright, deterministic simulation

WORKING PRINCIPLES

Uncertainty stays visible

Low-confidence input pauses or downgrades a judgment.

Recovery is product state

Permission, tracking loss, and interruption get explicit paths.

Evidence has boundaries

Engineering checks are never relabeled as user outcomes.

CURRENT FOCUS

Available for product engineering and applied computer vision work where software must understand the physical world.

SELECTED PRODUCT WORK / 2025-2026

Jiku Fitness

PRODUCT + ENGINEERING

Built a browser training loop for boxing, push-ups, and pull-ups around explicit play, rest, completion, results, confidence, and tracking-loss states.

84% TARGET OCCUPANCY ACROSS 347 RECORDED-REPLAY SAMPLES / 8 DETERMINISTIC BOXING TESTS

AR Baseball

GAME + POSE SYSTEMS

Built a camera-controlled three-inning baseball game with readiness calibration, fresh-frame swing recognition, fielding, runners, rival scoring, and bounded completion.

250K SEEDED SIMULATIONS / RECORDED-CAMERA PRODUCTION GATE / COMPLETE THREE-INNING LOOP

Bio-Tactical Edge

RESEARCH + PRODUCT ARCHITECTURE

Designed a sports-intelligence workspace linking broadcast video, court calibration, player paths, synchronized 3D replay, counterfactual recovery, search, and report evidence.

77 AUTOMATED TESTS / PUBLIC SOURCE-LINKED GOLDEN RALLY REMAINS THE NEXT ACCEPTANCE ARTIFACT

ADDITIONAL SYSTEMS SHIPPED

Jiku Tennis	body-tracked rallies and five-rival progression	33 TESTS
Form: Pickleball	pose, ball, and audio coaching workflows	695 TESTS
ASL Hero	self-hosted hand tracking and learned fallback	DECISION-PATH TESTS
AR Slicer	independent camera layer and thermal-aware quality	RECOVERY GATE
Orbitap	seeded courses and shareable ghost runs	PWA BUILD

ENGINEERING VALIDATION SNAPSHOT

250k BASEBALL SIMULATIONS	695 COACH TESTS	5 CAMERA PRODUCTS
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WHAT I CAN DO FOR YOUR TEAM

Prototype the hard input Turn gesture, pose, video, or physical motion into a legible interaction loop.	Make it trustworthy Design confidence, failure, privacy, and recovery as product states rather than edge cases.	Ship with evidence Use replay, simulation, browser gates, and honest metrics to protect the complete journey.
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AVAILABLE NOW

Product engineering / Applied CV / Interaction systems